

Sample Size Planning

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```
library(tidyverse)
library(brms)
library(tidybayes)
```

Why I am writing this notebook

I keep running into the same question, from students and from myself: “*I only managed to sample 12 ponds/streams/plots this season, is that enough?*” Or the opposite worry, after collecting the data: “*I found no effect, does that mean there isn’t one?*”

Both questions are really about the same thing: how much can the number of observations you collect distort what you learn from a model? I want to show you this directly, rather than just telling you “more data is better.” So in this notebook I simulate a very simple ecological relationship whose true parameters I fix myself, then I repeatedly draw samples of different sizes from that world and fit a Bayesian linear regression with **brms** to each one. Since I know the ground truth (I made it up), I can watch, sample size by sample size, how close my estimates get to it, how wide my uncertainty is, and how often I would draw the wrong conclusion if I only had one dataset in front of me — which, in real life, is exactly the situation you are always in.

This is not an abstract exercise. If you design a field campaign, a mesocosm experiment, or a monitoring programme, you are implicitly making a sample-size decision, whether or not you ever compute it explicitly. I would rather you make that decision with some idea of its consequences.

I am basing the simulation workflow on two blog posts I like a lot, and that I recommend you read directly if this notebook makes you want to go deeper:

- A. Solomon Kurz, [*Bayesian power analysis: Part I. Prepare to reject \$H_0\$ with simulation*](#)

- Tyson S. Barrett, *Bayesian Power Analysis with data.table, tidyverse, and brms*, which replicates and extends Kurz’s approach and adds power curves across several sample sizes, which is exactly the part I lean on most below.

Both posts work with a two-group comparison (treatment vs. control). I am going to adapt the same logic to a case you will run into constantly in ecology: a simple linear regression between two continuous variables.

i Note

A word on vocabulary before we start:

Power is a frequentist concept: the probability that, if the true effect is what you assume it is, your analysis would correctly reject the null hypothesis. It is not a fully “Bayesian” quantity, since Bayesians do not test null hypotheses in the classical sense. But I can still ask an analogous, useful question: *given the effect size I expect, how often would my 95% credible interval for the slope exclude zero?* That is the quantity I am going to simulate here, following Kurz and Barrett. It is a pragmatic, simulation-based way to reason about sample size, not a philosophical statement about what a Bayesian “should” care about.

The ecological scenario

Let’s say you are interested in eutrophication, and specifically in how total phosphorus (TP) concentration in a lake predicts chlorophyll-a concentration (a common proxy for phytoplankton biomass). This is one of the best-established relationships in limnology, and a good one to use here because you already have strong prior expectations about its sign and rough magnitude.

I will work on the log-log scale, which is standard for this relationship and which also conveniently turns things into a simple linear model:

$$\log(\text{chl-}a_i) = \beta_0 + \beta_1 \log(\text{TP}_i) + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma)$$

To keep the simulation clean, I standardize both the predictor and the outcome (mean 0, SD 1) before simulating. This means I can talk about β_1 directly as a standardized effect size, in the same spirit as Kurz’s and Barrett’s Cohen’s $d = 0.5$ example. I will fix:

- $\beta_0 = 0$ (intercept, arbitrary since both variables are standardized)
- $\beta_1 = 0.35$ (a moderate, ecologically plausible standardized effect of TP on chlorophyll-a)
- $\sigma = 1$ (residual/process variance, standardized)

I picked $\beta_1 = 0.35$ on purpose: it is a realistic, not-huge effect. Real ecological signals are rarely as clean as a textbook example, and moderate effects are exactly the ones that are easiest to miss with too little data — which is the whole point of this notebook.

```
# the "true" world I am simulating from
beta0_true <- 0
beta1_true <- 0.35
sigma_true <- 1
```

Dry run: what does one dataset look like?

Before I jump into simulating many datasets, I want you to see a single one, the same way Kurz starts with a “dry run.” Let’s simulate $n = 30$ lakes.

```
sim_d <- function(seed, n) {
  set.seed(seed)

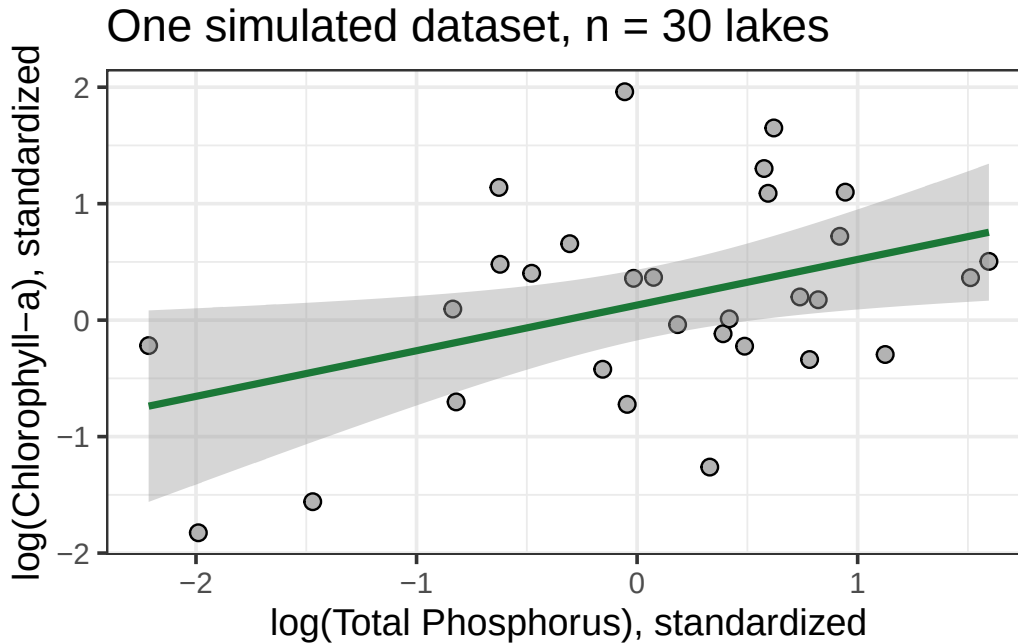
  x <- rnorm(n, mean = 0, sd = 1)          # standardized log(TP)
  y <- rnorm(n, mean = beta0_true + beta1_true * x, sd = sigma_true) # standardized log(chl.)

  tibble(lake_id = 1:n, log_TP = x, log_chla = y)
}

d30 <- sim_d(seed = 1, n = 30)
head(d30)
```

```
# A tibble: 6 x 3
  lake_id log_TP log_chla
  <int>   <dbl>   <dbl>
1     1 -0.626    1.14
2     2  0.184   -0.0385
3     3 -0.836    0.0952
4     4  1.60     0.505
5     5  0.330   -1.26
6     6 -0.820   -0.702
```

```
ggplot(d30, aes(x = log_TP, y = log_chla)) +
  geom_point(shape = 21, size = 2.5, fill = "grey70") +
  geom_smooth(method = "lm", formula = y ~ x, color = "#1B7837", se = TRUE) +
  labs(x = "log(Total Phosphorus), standardized",
       y = "log(Chlorophyll-a), standardized",
       title = "One simulated dataset, n = 30 lakes") +
  theme_bw(base_size = 14)
```



Now let's fit a Bayesian regression to it. I use weakly-regularizing priors, in the same spirit Kurz recommends: informative enough to keep the sampler in a sensible region, but nowhere near as informative as what I actually simulated the data from (I want the model to have to "discover" the effect from the data, not have me hand it the answer via the priors).

```
fit_base <- brm(  
  formula = log_chla ~ 1 + log_TP,  
  family   = gaussian,  
  data     = d30,  
  prior    = c(prior(normal(0, 1), class = Intercept),  
               prior(normal(0, 1), class = b),  
               prior(exponential(1), class = sigma)),  
  iter     = 2000, warmup = 1000, chains = 4, cores = 4,  
  seed     = 1,  
  backend  = "cmdstanr"  
)
```

Running MCMC with 4 parallel chains...

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Chain 1 Iteration:   100 / 2000 [ 5%] (Warmup)  
Chain 1 Iteration:   200 / 2000 [10%] (Warmup)  
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Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.

```

Chain 3 finished in 0.0 seconds.

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.2 seconds.

```
print(fit_base)
```

```
Family: gaussian
Links: mu = identity
Formula: log_chla ~ 1 + log_TP
Data: d30 (Number of observations: 30)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.13	0.15	-0.17	0.43	1.00	3472	2432
log_TP	0.38	0.17	0.05	0.72	1.00	3429	2642

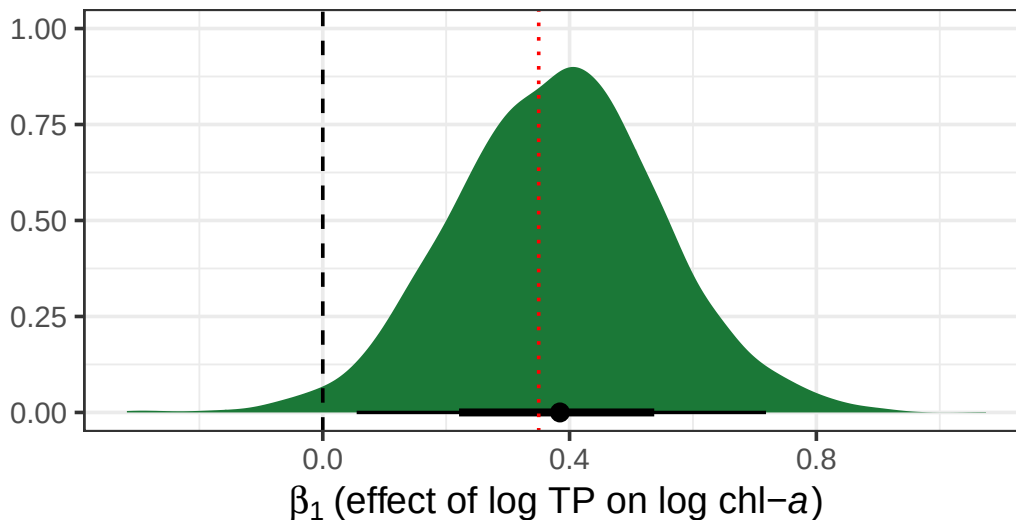
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.83	0.12	0.65	1.09	1.00	3096	2656

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
fit_base |>
  spread_draws(b_log_TP) |>
  ggplot(aes(x = b_log_TP)) +
  stat_halfeye(point_interval = median_qi, .width = c(.66, .95), fill = "#1B7837") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = beta1_true, linetype = "dotted", color = "red") +
  labs(x = expression(beta[1]~"(effect of log TP on log chl-"*italic(a)*"")),
       y = NULL,
       title = "Posterior for the TP -> chlorophyll-a slope, n = 30",
       subtitle = "Dashed line = zero. Dotted red line = the true value I simulated from.") +
  theme_bw(base_size = 14)
```

Posterior for the TP → chlorophyll-a slope, $n = 30$
Dashed line = zero. Dotted red line = the true value I simulated



With $n = 30$, the 95% credible interval already tends to exclude zero, and the posterior mean is somewhere in the neighborhood of the true 0.35 I set. But “somewhere in the neighborhood” is doing a lot of work in that sentence. Let’s see what happens if I repeat this many times, and at different sample sizes.

The full simulation: many datasets, many sample sizes

I am following Barrett’s approach here almost exactly, because it is the cleanest one for what I want: wrap data simulation, model fitting, and parameter extraction into a single function, then map it over a grid of seeds and sample sizes. I reuse `fit_base` with `update()` rather than recompiling the Stan model from scratch every time, which is what makes this fast enough to run inside a notebook.

```
sim_d_and_fit <- function(seed, n) {  
  
  d <- sim_d(seed = seed, n = n)  
  
  fit_i <- update(fit_base, newdata = d, seed = seed)  
  
  fit_i |>  
    fixef() |>  
    as.data.frame() |>
```



```

  rownames_to_column("parameter") |>
  filter(parameter == "log_TP")
}

```

I will explore six sample sizes, from quite small to reasonably large for a typical field campaign: 10, 20, 30, 50, 80, and 150 lakes. For each sample size I run `n_sim` independent simulated “studies.” I am keeping `n_sim` modest (50) so the notebook renders in a reasonable time on my laptop — if you were doing this for real, before applying for funding or committing to a field season, I would push this to 500 or 1000 simulations per sample size.

```

n_sim <- 50
sample_sizes <- c(10, 20, 30, 50, 80, 150)

grid <- expand_grid(n = sample_sizes, seed = 1:n_sim) |>
  mutate(seed = row_number()) # unique seed per simulated dataset

sims <- grid |>
  mutate(b1 = map2(seed, n, sim_d_and_fit)) |>
  unnest(b1)

```

Running MCMC with 4 sequential chains...

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Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
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Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

```
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
```



```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```

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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```



```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

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Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

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Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...


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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

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Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

```

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)

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Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

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Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

```

Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)


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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

```

```

Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

```

Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

```



```
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
```

Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)

```

Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

```

```

Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

```

Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

```

Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)


```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

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```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
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Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
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Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)

```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```



```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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```

Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```



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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

```
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
```



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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

```

```

Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)

```

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)

```

Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

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Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

```

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)

```

Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)

```

Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
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Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
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Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
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Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)


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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)


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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

```

Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
```

```

Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

```



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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

```

```

Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

```

Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)

```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

```

```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
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Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
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Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
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Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
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Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)

```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```


All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```

```

Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

```

```

Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```

```

Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```



```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

```

Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

```

Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

```

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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
```

```

Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)


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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

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```
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

```

```
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

```

Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

```


Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)

```

Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)

```

```

Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
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Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)


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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
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```

Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

```

Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
```

```

Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

```

```

Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

```



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Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

```

Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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```
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
```

```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

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Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```


Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

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```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```



```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

```
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
```

```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```


Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```

```

Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.


```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
```

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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)

```

```

Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```



```

Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```

Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
```

```

Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

```

```
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
```

```

Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

```

```

Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

```

```
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

```

Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)

```

```

Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

```



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Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

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Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
```

```

Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

```

```

Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)

```

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Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)


```

Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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```
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
```

```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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```

Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)

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Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

```
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
```



```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```

```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```

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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

```

Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```



```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

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Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
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Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
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Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
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Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

```

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Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)

```

```

Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

```

```

Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

```

Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

```

Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

```

Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

```
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

```
All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.
```

```
Running MCMC with 4 sequential chains...
```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

```

Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
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Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
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Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)


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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)

```

```

Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

```

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Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.5 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

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Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)

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Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

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Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

```

Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)


```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

```

```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```

Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
```

```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```



```

Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```



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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

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Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

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Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

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Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

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Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

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Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

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Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

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Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
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Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)


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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)


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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

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Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

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Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

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Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

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Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)

```

Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

```



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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

```

Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

```

```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

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```

Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

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Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

```

```
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
```

```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

```

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Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

```

```

Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```


All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
```

```

Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)

```

Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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```

Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```



```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

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Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.

```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)

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Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)


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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

```

Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)

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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)

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Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

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Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

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Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)

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Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

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Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

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Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
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Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
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Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

```

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)

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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

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Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)

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Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

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Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

```



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Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)

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Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
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Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

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Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```


Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)

```

```

Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)

```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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```

Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

```

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Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)

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Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)

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Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)

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Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)

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Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)

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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)

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Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)

```

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Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
```

```

Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.


```

Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration:  1800 / 2000 [ 90%] (Sampling)

```

```

Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)

```

```

Chain 1 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

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Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)

```



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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)

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Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
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Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

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Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)

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```
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
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Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
```



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Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)

```

Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)

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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

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Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)

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Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)

```

```

Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)

```


Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration:  1800 / 2000 [ 90%] (Sampling)

```

```

Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

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Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)

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Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

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Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)

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Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)

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Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
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Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)

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Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

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Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)

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Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
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Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
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Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

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Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)

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Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)

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Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)

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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

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Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)

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Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)

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Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)

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Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.

```

```

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)

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Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

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Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)

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Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)

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Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)

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Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

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All 4 chains finished successfully.
 Mean chain execution time: 0.0 seconds.
 Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)

```

```

Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)

```

```

Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)


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Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)

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Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)

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Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)

```

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Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

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All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

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```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)

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Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)

```

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Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)

```

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
```

```

Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)

```



```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
```

```

Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)

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Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:  500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:  600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:  700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration:  800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

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Running MCMC with 4 sequential chains...

```

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:  300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%] (Warmup)

```

```

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)

```

```

Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
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Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
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Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 2 finished in 0.0 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
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Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
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Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

Running MCMC with 4 sequential chains...

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Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
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Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
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```



```

Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.0 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
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Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.0 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
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Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.0 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
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Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
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Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.0 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.6 seconds.

```

```

sims <- sims |>
  mutate(powerred = if_else(Q2.5 > 0, 1, 0))

head(sims)

```

```

# A tibble: 6 x 8
      n seed parameter Estimate Est.Error  Q2.5 Q97.5 powerred
  <dbl> <int> <chr>      <dbl>    <dbl>  <dbl> <dbl>    <dbl>
1     10     1 log_TP    -0.146    0.416 -0.990 0.651      0

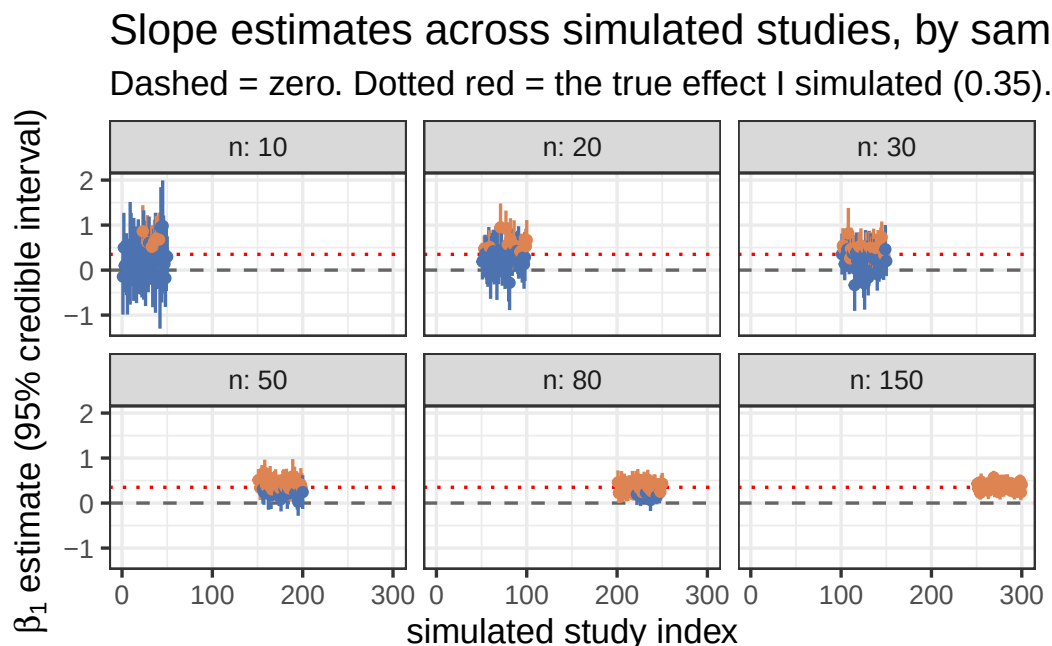
```

2	10	2 log_TP	0.499	0.398	-0.308	1.27	0
3	10	3 log_TP	0.101	0.304	-0.497	0.704	0
4	10	4 log_TP	0.0866	0.136	-0.170	0.361	0
5	10	5 log_TP	0.151	0.344	-0.559	0.821	0
6	10	6 log_TP	-0.196	0.400	-0.988	0.602	0

What sample size buys you: less noise around the truth

The first thing I want you to see is simply how the cloud of slope estimates behaves as n grows. Each point below is one simulated “study” — one imaginary field season with a given number of lakes — with its 95% credible interval. Points are colored by whether the interval excludes zero (yellow) or not (blue), following Barrett’s plotting convention.

```
sims |>
  ggplot(aes(x = seed, y = Estimate, ymin = Q2.5, ymax = Q97.5, color = factor(powered))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  geom_hline(yintercept = beta1_true, linetype = "dotted", color = "red") +
  geom_pointrange(fatten = 1) +
  scale_color_manual(values = c("0" = "#4C72B0", "1" = "#DD8452")) +
  labs(x = "simulated study index",
       y = expression(beta[1]~"estimate (95% credible interval)"),
       title = "Slope estimates across simulated studies, by sample size",
       subtitle = "Dashed = zero. Dotted red = the true effect I simulated (0.35).") +
  facet_wrap(~ n, nrow = 2, labeller = label_both) +
  theme_bw(base_size = 13) +
  theme(legend.position = "none")
```



Notice three things happening together as you move from $n = 10$ to $n = 150$ panels:

1. **The point estimates themselves get less scattered.** At $n = 10$, some “studies” would have told you the effect is close to zero, or even negative, purely by bad luck of the draw, even though I know the true effect is a solid positive 0.35. At $n = 150$, almost every simulated study lands close to the true value.
2. **The intervals shrink.** This is precision, in Kurz’s sense from the second post in his series: a narrower interval is a more informative one, independent of whether it happens to exclude zero.
3. **More intervals turn yellow (exclude zero).** This is power in the classical sense: with more data, you are more reliably able to detect an effect that is really there.

The power curve

Now let’s summarize the “yellow vs. blue” proportion at each sample size into a single number: my simulation-based estimate of statistical power, exactly as Barrett does for his power-curve figure.

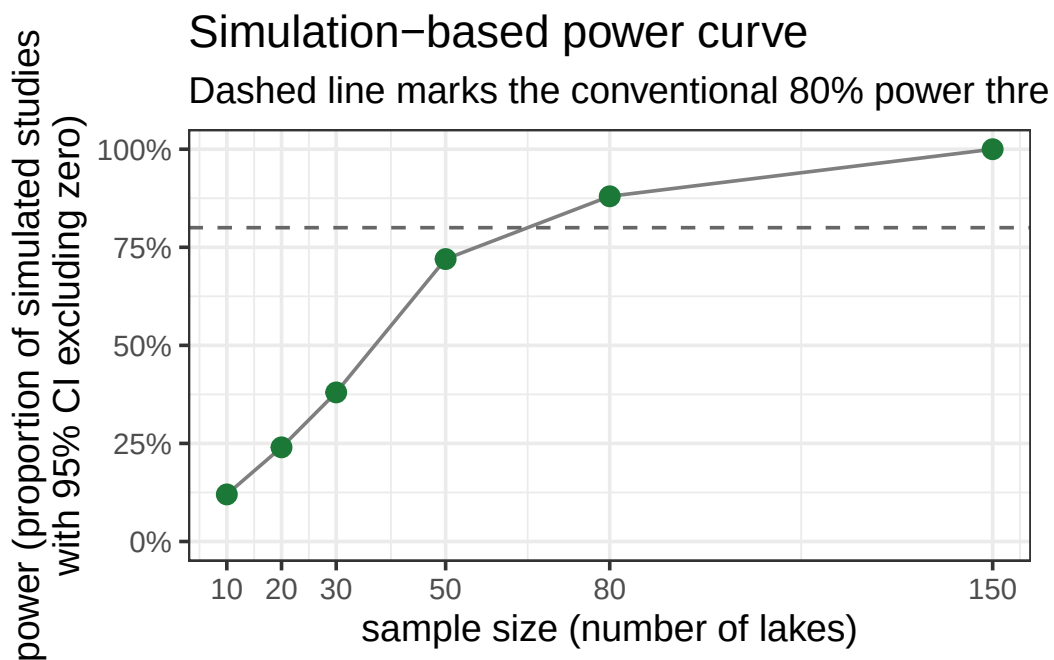
```
power_curve <- sims |>
  group_by(n) |>
  summarise(power = mean(power), .groups = "drop")

power_curve
```

```
# A tibble: 6 x 2
```

```
      n power
  <dbl> <dbl>
1     10  0.12
2     20  0.24
3     30  0.38
4     50  0.72
5     80  0.88
6    150  1.00
```

```
ggplot(power_curve, aes(x = n, y = power)) +
  geom_line(color = "grey50") +
  geom_point(size = 3, color = "#1B7837") +
  geom_hline(yintercept = 0.8, linetype = "dashed", color = "grey40") +
  scale_y_continuous(limits = c(0, 1), labels = scales::percent) +
  scale_x_continuous(breaks = sample_sizes) +
  labs(x = "sample size (number of lakes)",
       y = "power (proportion of simulated studies\nwith 95% CI excluding zero)",
       title = "Simulation-based power curve",
       subtitle = "Dashed line marks the conventional 80% power threshold") +
  theme_bw(base_size = 14)
```



! Take-home message

With only `n_sim = 50` simulations per sample size, this curve is itself a bit noisy — you would see it wiggle a little if you reran the whole notebook with a different starting seed. But the overall shape is exactly what you should expect, and is the reason sample-size planning matters: for the effect size and priors I specified here, you would need something in the range of 30-50 lakes to have a reasonable (~80%) chance of confidently detecting a moderate TP -> chlorophyll-a relationship. Below that, you are essentially flipping a weighted coin: sometimes you will get a clean, “significant” result, and sometimes you will not, even though the underlying ecological relationship never changed between simulated studies.

Precision, not just power

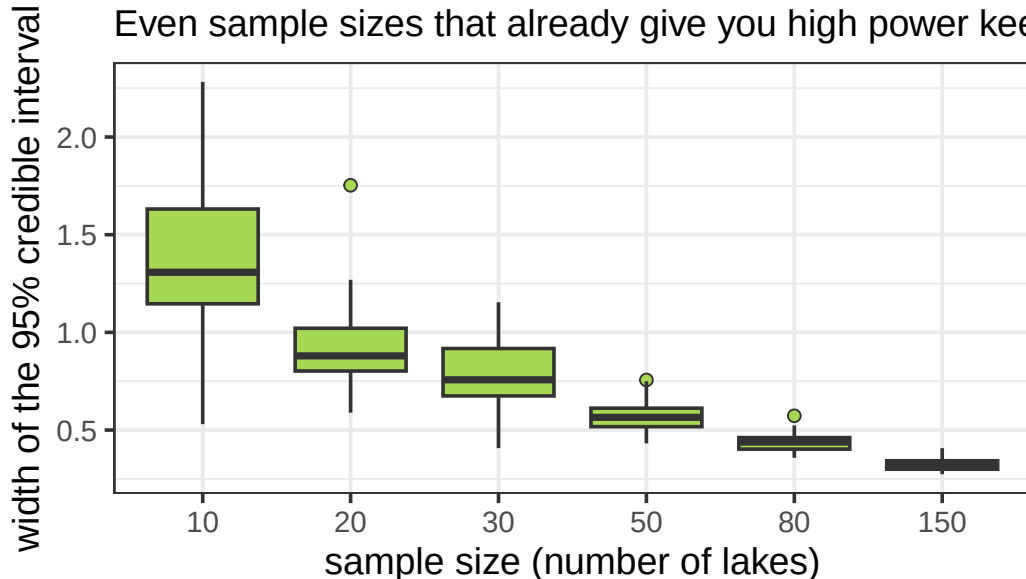
I don’t want you to walk away thinking the only thing sample size buys you is a yes/no answer about whether an interval excludes zero. That framing is useful, but it is also exactly the “significance-chasing” mindset that Kurz warns about in the sequel to the post I used here, on precision-oriented sample size planning. What sample size really buys you is a narrower, more trustworthy estimate of the *magnitude* of the effect, which is usually what you actually care about as an ecologist. A “significant” result from $n = 10$ is not more trustworthy than a “non-significant” one from $n = 150$; if anything, it is less so, because small, noisy samples that do happen to cross the significance threshold tend to overestimate the true effect size (sometimes called “the winner’s curse” or type M/S error in Gelman and Carlin’s terminology).

Let’s look directly at how interval width shrinks with n :

```
sims |>
  mutate(width = Q97.5 - Q2.5) |>
  ggplot(aes(x = factor(n), y = width)) +
  geom_boxplot(fill = "#A6D854", outlier.shape = 21) +
  labs(x = "sample size (number of lakes)",
       y = "width of the 95% credible interval",
       title = "Precision improves steadily with sample size",
       subtitle = "Even sample sizes that already give you high power keep getting more precise",
       theme_bw(base_size = 14)
```

Precision improves steadily with sample size

Even sample sizes that already give you high power keep



You can see that interval width keeps shrinking well past the point where power has already reached ~100%. If your goal is not just “detect that TP matters” but “estimate how much chlorophyll-a rises per unit increase in TP, precisely enough to inform a nutrient-reduction target,” you may need a considerably larger sample than the one that merely gets you past the power threshold.

How I would use this in practice

If you are planning your own field season or experiment, I would suggest you go through the same steps I just walked through, but with your own numbers plugged in:

1. Write down the simplest possible model of the relationship you expect (here: a single slope, on a sensible transformed/standardized scale).
2. Commit to a plausible effect size, ideally from a pilot study, a meta-analysis, or at minimum an honest, defensible guess, the same way I fixed $\beta_1 = 0.35$ above rather than pulling it out of thin air at the last minute.
3. Pick priors you could defend to a skeptical reviewer, not priors chosen to make your desired result more likely to reach significance.
4. Simulate datasets across a range of feasible sample sizes and fit your actual model to each one, using `update()` to keep it fast.
5. Look at both power and precision, not just power. Ask yourself what interval width you would actually need to answer your ecological question usefully, not just to get a star next to a p-value.

6. Remember that everything above is conditional on the effect size and priors you assumed. If the true effect is smaller than you guessed, your realized power will be lower than the curve above suggests. This is exactly why I like doing this via simulation instead of a one-line power formula: it forces you to be explicit about every assumption you are making, and it is trivial to rerun with a different guess and see how sensitive your conclusions are to it.

i Note

I kept this notebook to a single continuous predictor and a Gaussian likelihood on purpose, to keep the logic visible. The exact same workflow — simulate, fit with `update()`, extract, summarize — extends directly to more complex ecological models: Poisson or negative-binomial counts, hierarchical models with random effects for site or year, or the Gaussian Process models I explore in my [GP notebook](#). The simulation loop does not change; only the `sim_d()` and model-fitting steps do.